

SECTION 2: INSTALLATION

2.1 - SELECTION OF A BOILER

THE FOLLOWING FACTORS SHOULD BE GIVEN VERY CAREFUL CONSIDERATION IN SELECTION OF THE BOILER:

(1) SIZE OF THE BOILER: The Boiler must be of adequate capacity to handle the maximum hourly load of the particular application, plus radiation losses. It is considered good practice to install a Boiler of sufficient capacity to handle future expansion requirements or to allow space for an additional future Boiler. On critical or large installations, it is recommended to install two or more Boilers, each sufficient size to handle the minimum load so that the operation is not interrupted when one of the Boilers is being serviced. If the load is not definitely known, it is advisable to consult the boiler manufacturer or competent personnel to determine the load.

(2) APPROVED TYPE QUALITY BOILER:

The Boiler should be a quality, approved Boiler, manufactured and furnished in full compliance with the Safety Requirements of the A.S.M.E. Code. The Boiler selected should be safe, efficient and easy to maintain to provide economical and reliable service. The Parker Bent Water Tube is highly recommended as it provides one of the safest, most economical and long life systems that is easy and inexpensive to maintain.

(3) TRIM AND CONTROLS:

The Boiler should be equipped with fully automatic controls to provide automatic and safe operation. All Trim and Controls should comply with Local Code, the State Safety Orders, the A.S.M.E. Codes and Underwriters' Laboratories. A flame safeguard, operating control, high limit, low water cutoff and dual safety shutoff fuel valves are considered essential on all Boilers.

(4) PROPER BOILER SPECIFICATIONS:

The complete Boiler Specifications should be thoroughly reviewed for the proper temperature and working pressure required the most economical fuel available and proper electrical requirements. Insurance requirements for the particular installation should be reviewed to determine if F M (Factory Mutual) or GE GAP Code Trim is required.

2.2 - INSTALLATION AND OPERATING PERMITS

Before installing the Boiler be certain that all Permits for the installation and operation of the boiler have been taken out and installation conditions approved by the Building Inspection Jurisdiction. Permits are usually required on the Boiler and Utilities including Gas, Electrical, Water, Vent and Plumbing Connections. Some States and Cities require a Field Inspection by a State or National Board Commissioned Boiler Inspector and a "Permit to Operate" issued before the Boiler is placed in service. Heavy Penalties and Unsafe Conditions may result when these Permits and Inspections are not obtained from the Inspection Jurisdiction. If Local Codes or Insurance Carrier require additional or special controls, the boiler manufacturer should be consulted so that the equipment can be furnished as required.

Additionally, many local air pollution districts require permitting of boilers and a Low NOx boiler may be required. In many air districts, certification to SCAQMD Rule 1146.2 is required.

Parker Low NOx units up to 2,000,000 BTUH input are SCAQMD certified to Rule 1146.2 and may be installed in many districts. Check with the local APCD, your sales agent and factory to confirm Low NOx and permit requirements.

2.3 - BOILER ROOM AREA

Consult with Local Codes for Boiler Room and Fire Resistant requirements. In most building occupancies, a boiler room is required for boilers over 400,000 BTUH input.

The Standard Boiler is not recommended for outside installation without providing the required protection from weather and drafts. Optional weather protection or outdoor trim is available from the factory and the boiler must be ordered this way.

An approved floor sump or drain must be provided in the Boiler area to accommodate safety relief valves, boiler drains and leaks from any source. Floors should be properly sealed, planned, and graded to the floor sump or drain to prevent water damage.

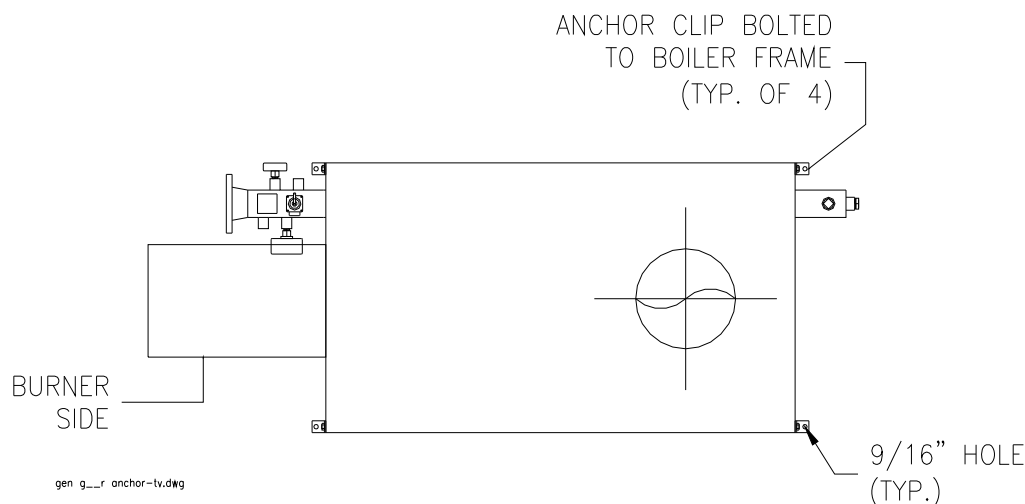
When Boiler Rooms are necessary, two access doors are recommended and normally required on rooms larger than 500 sq.ft. The larger door should be adequate to permit the Boiler to be moved in and out without disassembly. Location of the Boiler Room should be selected as conveniently near the load as possible with all installation factors considered particularly, the boiler vent and outside air source.

Do not install the boiler in any location where flammable liquids or vapors are likely to be present or drawn. Keep the area around the boiler free of combustible materials, gasoline, paint, thinner, solvents and other flammable liquids and vapors.

2.4 - BOILER BASE / ANCHORING

BOILER MUST BE INSTALLED ON A NON-COMBUSTIBLE FLOOR preferably of concrete and properly designed to support the required weight and to protect against any Fire Hazard. Combustible floors require an Approved Fire-Proof Base, consult Manufacturer. Do not install on wood, carpet, vinyl tiles, black-top or on concrete with wood or electrical conduit imbedded or directly underneath without consulting Manufacturer for an approved base.

The boiler may be shimmed level if required. Four (4) anchor points are provided at the boiler base and optional anchor clips bolt to the cabinet frame. These anchor clips can be bolted to a concrete base. A concrete housekeeping pad for the boiler is recommended but not required.



2.5 – CLEARANCES

CLEARANCE TABLE

		Thermal Fluid Heater	Low NOx Boilers 115V-1Ph	Power Burner Models
	Model	Minimum listed cabinet clearances to combustible construction	Recommended clearance for service / electrical	Recommended Clearance for service / electrical
Top		48"	X	X
Front		48"		
Sides		48"		
Back		48"	14" **	14" **
Vent Connector		18"	18"	18"

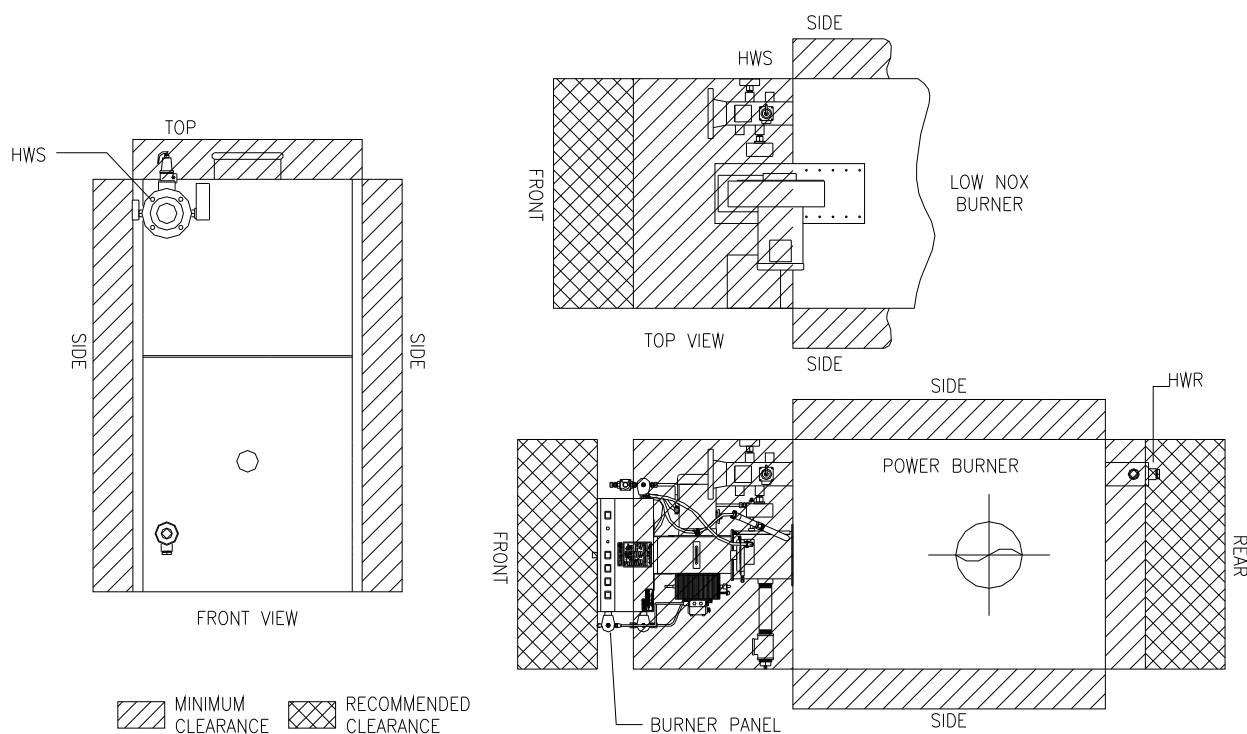
NOTE: **Some States require 36" clearance around boiler (Arizona, Nevada, Utah and others). Confirm space and clearance requirements before installation.

*Most jurisdictions require 30" clearance in front of electrical panel for 115V service and 36" or 48" for higher voltages with 3 Phase service.

HWS - Hot water piping exits front of boiler.

THERMAL FLUID HEATER REQUIRE 48" ALL AROUND

HWR - Hot Water piping enters back of boiler.



TOP VIEW

gen g...r With clearances.dwg

2.6 - COMBUSTION AIR SUPPLY

The Boiler requires fresh air for safe operation and must be installed so there are provisions for adequate combustion and ventilation air. In confined rooms (rooms whose volume is less than 50 cu.ft. per 1000 BTU aggregate input) or unconfined boiler rooms of tight construction, two permanent air openings shall be provided, one within 12" of the top and one within 12" of the bottom of the boiler room. Outside air supply sizes must be determined considering all fuel fired appliances in the room.

AIR SUPPLY FROM OUTSIDE BUILDING: If the boiler room is confined or the building is of tight construction, openings shall be provided that communicate freely with outdoors. For exterior wall openings or vertical ducts, each opening shall have a minimum free area (after louver consideration) of 1 sq.in. per 4,000 BTUH input. Total free area should be 2 sq. in. per 4,000 BTUH input.

Example:

2,000,000 BTUH Input / 4,000 BTUH Input = 500 sq.in. free area high and low

Total = 1,000 sq. in. free area

(For 75% free area louver requires 1,000 sq.in. / .75 = 1,334 sq.in. total opening)

When connecting with the outdoors through horizontal ducts, each opening shall have a minimum free area of 1 sq.in. per 2,000 BTUH input.

AIR SUPPLY FROM INSIDE BUILDING FOR UNCONFINED ROOMS: The two openings should have a total free area of 2 sq.in. for each 1000 BTU of input communicating directly with an unconfined space with adequate infiltration.

Louvers typically have from 50% to 75% free area, thus openings must be larger.

In hot climates additional ventilation should be provided to maintain reasonable temperatures in boiler room. Always leave adequate clearance around the base and legs of the Boiler so that air freely enters the combustion area. Air supply should be provided from a non-hazardous and clean source so that no flammable vapor, lint or dust is admitted to the boiler room. The Boiler air supply must be isolated from any source of corrosive chemical fumes such as perchlorethylene, chlorine, etc. Also take steps to minimize dust and lint in the area of the boiler.

The boiler room should always be neutral or slightly positive pressure 0.0" to +0.05" W.C. NEVER negative.

Exhaust fans are not recommended in a boiler room. A relief type louver or ceiling gravity ventilator or openings can help remove heat from the room or a small supply air fan can be used.

The Water Wall Boiler is a sealed combustion unit and outside air can be ducted directly to the blower. Consult factory for limitations. Additionally, engineered combustion air supply systems may be utilized if properly designed and approved by the building official. Consult factory for special conditions.

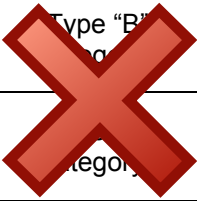
Low NOx units are provided with an air inlet filter. The filter should be kept clean at all times. Always keep air inlet screens and louvers clean.

2.7 - COMBUSTION VENTS

The proper boiler venting system is required to assure the necessary draft for efficient combustion and to develop a positive flow adequate to convey all products of combustion safely to outside.

The vent system must be in accordance with the National Fuel Gas Code/NFPA 54 ANSI Z222.3, Part 7, Venting of Equipment or Applicable provisions of Local Building Codes.

Parker Gas Fired Water Wall Boilers are fan assisted gas fired appliances and may be vented per the vent tables.

MODEL	VENT TYPE REQUIRED	
	Section IV Boilers up to 250°F	Section I Boilers over 250°F
Low NOx Models Natural Gas or LPG Gas Fired		Chimney
Power Gas Fired Units with Powerflame Burner Gas Fired Burners		Chimney
THERMAL FLUID HEATER, OIL FIRED UNITS	Chimney	Chimney

All venting must be in conformity with accepted engineering practices to the Manufacturer's minimum standards and in compliance with the Local Code in every respect. All Vent systems should be installed in a safe workmanlike manner, properly supported and joints secured to prevent leakage. Never drill holes or screw metal screws into double wall vent. **DO NOT SUPPORT THE WEIGHT OF THE VENT ON THE BOILER.**

A Barometric Damper and Single Wall Tee is often provided with the boiler. This Tee and Damper are normally fitted directly on top of the boiler. The barometric damper if provided must be installed in the same room as the boiler.

The Boiler Vent should preferably be circular and of metal. Do not reduce vent size or connect more than one appliance to a single vent unless in compliance with good safe accepted practices and with approval of the Building Inspection Authority. The Main Vent should run straight up from the Boiler and if horizontal runs are necessary, these should be as straight and direct as possible with a minimum rise of at least 1/4" per lineal foot. Long horizontal runs are not recommended and usually must not exceed 75% of the vertical height of the vent above the connector. Proper vent clearance must be provided from any combustible material, wall openings or exit way. In cold climates, outside vents must be properly insulated.

Do not terminate vents near windows, doors, ducts, air conditioning or air supply fans which might pick up flue products exhausted from the boiler room and return them into the building, causing a serious health hazard to occupants. An unrestricted full size vent cap should be installed on the top of the stack to protect from wind and rain.

When installation conditions require special vents or draft inducers, the design and safety controls should be approved by Competent Combustion Personnel. When a draft inducer is required, always install a suitable draft switch and wire this into the boiler control circuit to prevent the boiler from firing unless the draft is proven. If a draft inducer is used, the boiler requires a positive pressure type chimney from the outlet of the inducer to the termination. In planning the vent

installation, carefully review these Instructions and consult with the Building Inspection Authority on the Local Code for the Required Approval.

These are the Manufacturer's Recommended Minimum Standards but all installation materials and conditions must conform to the Local Code Requirements and good safe engineering practices for the particular installation.

TYPE B VENT SYSTEMS, CATEGORY I, GAS FIRED

These systems may be vented per the vent tables for a fan assisted Category I appliance. These tables show the vent required based on BTUH input, vertical and horizontal distances.

In most cases the vent size required will match the boiler output but for short vertical runs or large horizontal runs. A size increase may be required.

Type B vents should always be installed in accordance with the Manufacturer's Instructions. Listed Double Wall Vent is preferred and must be used in any attic, concealed space and for any combustible floor, ceiling or roof penetration. Vent connectors should be galvanized steel minimum 24 gauge thru 12" diameter and 22 gauge on larger diameters. Elbows used as vent connectors may be one gauge lighter. The Vent must terminate at least 2 feet above the roof or any part of the building within 10 feet. Single wall metal pipe vent is allowed as the main vent by the Local Inspection Jurisdiction. It should be a minimum 20 gauge galvanized material and used only for runs directly from the boiler area to the outside. Single wall metal pipe vent installed through a combustible roof or wall must be guarded at the point of penetration with the required ventilated thimble or combustible material cut away to provide not less than 18" clearance on all sides of the vent. When a size vent installed, no spillage should be observed at the draft control. A minimum draft of .02" W.C. negative pressure should be experienced below the barometric damper or at outlet collar. If the draft registers in excess of 0.06", Competent Combustion Personnel or Gas Company Engineer should be consulted for combustion test and adjustments to control draft for best efficiency. Proper start up is always required.

CHIMNEYS

ALL BOILERS LISTED FOR CHIMNEY MUST BE VENTED PER ALL CODES AND THE FOLLOWING: These Boilers may be classified as Residential-Type, Building Heating or Low Heat Appliances with flue gas temperatures below 600°F. A UL Listed Factory Built Chimney (Minimum Residential Type or Building Heating Appliance Type for Negative Pressure) should be installed in accordance with the Manufacturer's Instructions. Chimney connectors shall be galvanized steel minimum 22 GA thru 12" diameter, 20 GA thru 16" diameter, and 16 GA over 16" diameter. The chimney must terminate at least 3 feet above the roof and 2 feet above any part of the building within 10 feet. When a chimney extends through any story above that in which the boiler is located, including drop ceilings and crawl spaces, it should be enclosed within continuous non-combustible construction of the appropriate fire resistive rating with a minimum 12" clearance for access. If a single wall metal chimney is used it should be steel minimum of 10 GA. Support for this type chimney should be independent of building construction with the load transferred to the ground. Single wall metal chimney installed through a combustible roof or wall must be guarded at the point of passage with the required ventilated metal thimble or the combustible material cut away to provide not less than 18" clearance on all sides of the chimney. The Red Stops on the Barometric Damper should be removed on gas fired boilers but must remain in place on all oil and combination fired boilers.

When a Flue Gas Spillage Switch is furnished on gas fired boilers it is pre-wired but must be properly mounted above the damper as described on a separate bulletin. On all jobs, it is required to call in Competent Combustion Personnel to make a Complete Combustion Test for setting of draft to obtain best efficiency and safety and fill out ISCS sheet. Draft should be set for best combustion efficiency as low as possible without the presence of CO or Smoke in the flue gas.

With full size vent installed to proper height, a minimum draft of not less than .02 to .06" W.C. negative pressure should be experienced below the barometric damper. Start up to be performed per instructions.

2.8 – CONTROL VALVE VENTS

On certain Systems, the gas pressure regulator and other gas components require separate vent lines to outside atmosphere. These controls are marked when vents are required and they must be piped full size outside the building to a Safe Point of Discharge away from all personnel, ignition sources and air intakes. Gas pressure regulators above 1" size (above 3/8" size for medium or high pressure gas service) must be piped outside.

Controls requiring atmospheric vents such as gas pressure switches and gas pressure regulators that are not equipped with integral vent limiters, must be piped outdoors to a safe point of discharge. Do not vent into combustion chamber. A means must be provided at the termination point to prevent a stoppage of the line by foreign material, moisture, or insects. In the event of diaphragm rupture, gas will be discharged from vent.

Atmospheric vent lines must not be connected to any gas vent, bleed or relief line. Atmospheric vent lines, when manifolded, shall be connected to a common vent line having a cross sectional area not less than the area of the largest vent line plus 50 percent of the areas of all the additional vent lines.

2.9 – SAFETY RELIEF VALVE

The Safety Relief Valve discharge outlet must be piped full size independent of all other piping without any shut-off valves. Safety Relief Valve Drain may be larger than valve inlet size and must always be piped full size. Install piping with sufficient flexibility to allow for free expansion and properly support so there is no strain on the Safety Relief Valve Body and Piping. Pipe to a safe point of discharge to prevent any possibility of personal injury and within 18" from the floor or into an open receptacle protected by a Splash Shield. If discharge cannot be piped to a completely Safe Location in the boiler room, such discharge should be piped outside the room to a Safe Location. If piped upwardly a drain line should be provided at the low point to keep this line drained. Secure the piping so it cannot rise to cause personal injury when Safety Relief Valve discharges. Install a union near the Safety Relief Valve Outlet for convenience of changing valve when required.

Black Steel Schedule 40 pipe with malleable fittings is recommended.

2.10 – PIPING

Use ASME Code approved pipe, fittings and valves. The maximum temperature rating for malleable fittings is 500°F for Class 150 and 550°F for Class 300.

BOILER DRAIN: A Drain Valve must be installed on the lower header at the piping provided. Connect to a drain receptacle for draining Boiler or flushing when necessary. The Boiler is designed for forced system heating and should be flushed and drained only periodically when dirty. Any unnecessary loss of treated water from the Boiler System can be detrimental to the life of the Boiler and System piping unless carefully controlled.

WARNING: ALL PIPING MUST BE SECURED SO THAT IT CANNOT COME LOOSE DURING BOILER OPERATION. DO NOT SHOCK, BEND OR APPLY ANY STRAIN TO PIPING, CONTROLS AND VALVES.

2.11 – UTILITIES

(A) FUEL SUPPLY:

Do not use this boiler with any other fuel than the one listed on the nameplate. Provisions must be made for adequate fuel supply, proper size fuel lines, proper size gas meter and correct gas pressure. Refer to the correct Specification Sheet for the B.T.U. Input, gas pressure required and G.P.H. oil required. It is essential that the correct type fuel to specification and an adequate fuel supply be provided to assure efficient operation, proper boiler operating capacity and safety. All fuel piping must conform to the Local Code Requirements. Piping must be securely braced and fittings must always be clean and free of excessive pipe joint compound, Teflon tape and cutting and threading burrs and chips. Pressure test fuel piping in accordance with national and local codes. **Never pressure test a fuel line with the Boiler connected as this may cause damage.** Before firing the Boiler, be sure that all controls, valves and piping are tight and did not come loose in shipment. Use a soap-and-water solution to check for fuel leaks. Never use a match or open flame. Check that all test plugs on the fuel line are securely in place before firing and after making all adjustments. **ONLY QUALIFIED BURNER SERVICE PERSONNEL SHOULD START-UP, ADJUST OR SERVICE THIS EQUIPMENT.**

On Gas Fired Boilers one Upstream Gas Cock is furnished and an additional Gas Shutoff Cock (not furnished) is required to be installed near the Boiler. A drip leg should be installed at low point on the gas line just before connecting to the Boiler controls. Gas lines should be cleaned out before connecting to the Boiler.

Note that the gas service line may be required to be larger than the gas inlet connection. Refer to sizing tables. Install gas line per UPC.

Insure gas supply pressure to boiler matches nameplate value during static and full flow conditions.

(B) NATURAL GAS:

Natural Gas Burners are satisfactory for use on Natural Gas Fuel, 950 to 1150 B.T.U. content per cubic foot with .65 specific gravity. The Manufacturer should be notified of any change in pressure or fuel specifications that are not Standard so that the burner will be properly furnished for best operating efficiency and safety. The Gas Company should be notified before installation so that proper gas can be provided.

Standard gas pressure to boiler is 7-14" W.C. during static and full flow conditions.

A high inlet gas pressure option is available for 1-5 PSI during static and full flow conditions.

(C) LIQUID PETROLEUM GAS (L.P.G.):

The burner may be furnished for Liquid Petroleum Gas (Butane or Propane) firing when so specified. Provisions should be made with the Gas Supplier for the proper capacity, pressure requirements and pressure regulator when required.

The Local Requirements for installation of the tank and supply lines should be properly investigated prior to installation and fully conform to Code Requirements. The Manufacturer should be furnished with the Gas Specifications with the Order.

On LP Gas, a pressure 11-14" W.C. is standard at the gas inlet. The regulator furnished will reduce burner pressure for proper rate. A separate pilot line gas pressure regulator is furnished.

A high inlet pressure 1-5 PSI inlet is available at the gas inlet as an option.

(D) FUEL OIL:

Oil burners are standardly furnished for No. 2 light fuel oil, A.S.T.M. Specification D-396, with approximately 141,000 BTU content per gallon. Oil tank installation and connecting lines should be in full compliance with Local Code Requirements. The size of the oil supply tank should be to the advantage of the particular installation in considering the capacity for the most economical quantity purchase and of sufficient storage for the delivery service available.

A Two Pipe System (suction and return) is recommended on all installations and required on jobs with oil supply tank below burner level. A minimum 3/8" suction line is required up to 0-672 size, 1/2" suction line up to Model O-1008, 3/4" suction line size up to Model O-2640 as a minimum. Increase size of suction line if over 30' long. The return line may be one size smaller. On any suction line exceeding 50' or more than 10' lift, an auxiliary oil pump is required to keep continuous flow of oil circulating through line at all times while boiler is in use. An oil filter must be installed near the burner. If the oil supply inlet pressure is above 3 PSI, a special pressure regulating valve is required. The suction pressure at pump inlet shall never exceed 10" Hg. The fuel line connections at the burner should be flexible pipe or copper tubing for service convenience. For proper oil piping details, see the oil burner instructions. Do not use Teflon tape on oil piping.

The proper size oil nozzle(s) are furnished for the rated pressure of the burner. Competent trained service personnel should install oil gauge(s) as shown in burner manual to check pressure. Remove gauges when pressure settings are correct to prevent damage to the gauges.

On Gas/Oil Boilers remove pump coupling if oil cannot freely circulate thru oil pump if separate oil pump is not provided.

(E) ELECTRICAL:

Each Parker Hot Water Boiler is standardly furnished with the control circuit completely wired and factory fire tested. Field installation requires a main line disconnect switch(es). Electrician must read wiring diagram. Use Copper Conductors Only. The main line disconnect switch(es) should be of the circuit breaker or fused disconnect type. The Boiler must be grounded in accordance with the National Electric Code. Do not use gas piping as an electrical ground. All electrical connections should be furnished in complete accordance with the Local Code Requirements and must be in conduit. **ALWAYS TURN OFF ALL MAIN LINE DISCONNECT SWITCHES AND ALL OTHER SOURCES OF POWER BEFORE WORKING ON ANY ELECTRICAL CONTROLS.** The Electrical Service must match the Specifications of the Equipment:

Confirm voltage is correct before powering boiler, as damage may occur.

- (a). For models furnished for 115 Volt, 60 Hertz, 1 Phase for Boiler Controls adequate electrical supply for at least 15 Amps is recommended for the Boiler plus additional capacity as required to handle the Circulating Pump(s) if provided. Maximum size overcurrent device for boiler control circuit must not exceed 20 Amps.

Larger size boilers require 208/230/460 Volt, 60 Hertz, 3 Phase for Blower Motor. A control transformer for the 115 Volt boiler controls may be furnished where necessary. On all models where a separate oil pump is furnished, an additional line service may be required. **NOTE:** Special, higher elevations, etc. may change electrical requirements. Adequate electrical supply for at least 20 Amps is recommended for each boiler service plus additional capacity as required to handle the circulating pump- if provided.

(F) WATER SUPPLY FOR HOT WATER SYSTEM:

Provisions must be made for proper and adequate water supply and piping to conform to Local Code. Water makeup should be supplied to the System at the desired point. Be certain that the proper backflow device to conform to Local Code is supplied. On a direct water feed control, be certain that the city water pressure exceeds the Boiler operating pressure by a minimum of 10 PSI to assure positive water feed to the Boiler. Pressure rating of direct water feeders must be equal or greater than boiler safety relief valve setting.

2.12 – WATER TREATMENT

CLEANING & WATER TREATMENT INSTRUCTIONS **CLOSED HOT WATER BOILER HEATING SYSTEMS**

I. GENERAL DESCRIPTION ON A CLOSED HOT WATER HEATING SYSTEM

A closed hot water system is one where the hot water boiler, piping and the heat exchange or radiation units are filled with water and tightly sealed from leaks and outside air. The expansion tank maintains the proper proportion of air and water and serves as an area for water expansion during the heating cycle. An ideal closed system is one with no leaks and where no air or water are added or released. If this type of system is initially clean, properly conditioned and maintained against leaks, it will provide efficient and long life service.

The water in the system should never be drained unless repairs, changes in the system or dirty system water make it necessary. Changing of the water or adding untreated water in the system (which contains lime and sediment that increases the corrosion and scale possibility), upsets the balance of air and water.

Water never loses its heating transfer efficiency unless contaminated by foreign matter. Water in a closed heating system improves with age and should not be changed unless absolutely necessary.

II. NEW INSTALLATION OF CLOSED HOT WATER HEATING SYSTEMS

Every closed hot water system should be properly cleaned when the installation is completed and before putting the system in operation. This includes the boiler and the entire piping system. During installation, most systems become contaminated by pipe dope, thread cutting oils, soldering flux, rust preventives or dirt in the pipe and fittings. There are frequently sufficient quantities of these contaminants that break down chemically causing gas formation and acid in the system water. Once the system is contaminated, deterioration continues, leaks develop and water losses increase which may cause serious damage due to the scaling or corrosion of the boiler. To eliminate this potential hazard, the system must be initially cleaned and water conditioned when placed in operation. (See Page 2, Paragraphs IV & V). During the first month, supplemental inhibitors such as sodium nitrate and borate or phosphates and silicates are recommended to build a thin film coating on the tubes and piping. Protect the system from slime or organic growth.

We recommend operating the system normally with the make-up water valve closed so system leaks will result in a low water condition. Thus operational personnel will be quickly made aware of any system leaks and they can effect repairs.

III. EXISTING CLOSED HOT WATER HEATING SYSTEMS

Existing closed hot water systems should not be drained or cleaned, unless they become dirty or if it is necessary to do so for repairs or changes. Flushing and draining necessitates adding of make-up water which may cause corrosion, scale and other problems.

The following are the indications that an existing system needs repair or cleaning:

1. Visible pump or other system component leaking.
2. Scale formation on the outside of pipe, fittings, valves or sight glass openings.
3. System in need of constant make-up water. This can be determined by closing water make-up valve and monitoring sight glass on expansion tank.
4. Obviously discolored, murky, dirty, or rusty water.
5. Vent gases at high points in the system. These will ignite and burn with an almost invisible blue flame.
6. A pH or alkalinity test with a pH reading below 7. (Below 7 pH indicates that the system water is acid. The recommended pH level is between 8 to 9). A system that is acid will usually have one or more of the following symptoms:
 - a. Gas formations (air troubles).
 - b. Pump seal and gland problems.
 - c. Air vent valves sticking or leaking.
 - d. Frequent relief valve operation.
 - e. Pipe joint leaking.

CLEANING & WATER TREATMENT INSTRUCTIONS CLOSED HOT WATER BOILER HEATING SYSTEMS

IV. CLEANING OF A HOT WATER SYSTEM

Cleaning a hot water system is neither difficult nor expensive when the proper procedure is followed. For proper cleaning use Parker Clean-1 chemical or equal as recommended for such purpose and according to each supplier's procedure:

1. For new installations, add a one pound can of Parker Clean-1 for each 50 gallons of water in the system. Dissolve one pint of Clean-1 in one gallon of water and pour into air space of expansion tank. For existing installations, use Parker Solve-1 in the same manner.
2. Fill, vent and circulate the system with the cleaning solution at approximately 160° - 175° for 2 to 4 hours.
3. Completely drain the cleaning solution from the boiler and system and flush with clean water to completely remove all cleaning solutions.
4. Fill and vent the system again and fire the boiler to 150° for 30 minutes to one hour.
5. Completely drain the rinse water and flush with water until the drain water is clear and thoroughly clean.
6. Refill the system to the proper water level. Vent air in expansion tank and resume normal operation.

V. WATER TREATMENT FOR A HOT WATER HEATING SYSTEM

The system must be thoroughly cleaned before effective control can be maintained. Once the system is cleaned, a liquid water treatment is recommended to prevent scale, corrosion and foaming. A liquid treatment should be used to prevent damage to the pump seals. Parker Boiler Control or equal should be used to properly maintain the system.

1. Shake well before using.
2. Add one quart minimum of Parker Boiler Control for each 50 gallons of water in the system. This liquid should be poured directly into the expansion tank of the system. Parker Boiler Control will raise the pH to the safe recommended level of 8.0 to 9.0.
3. Carefully maintain the system to eliminate the loss of water due to leakage in pipes, valves, fittings, seals and packing glands.
4. A clean, neutral hot water system need never be drained except in an emergency or due to necessary servicing of the equipment. If the system is drained, Parker Boiler Control should be added as above instructions 1-2.
5. If no make-up water is added, additional treatment is not necessary unless determined from an analysis of the boiler and system water. If make-up water is added, all leaks should be immediately repaired and Parker Boiler Control added, use 1/2 pint for each 15 gallons of make-up water.

Notes: On systems using ethylene glycols, an inhibited non-automotive grade should be used (the recommended pH level is 8-10). When using ethylene glycols, a supplemental inhibitor package of sodium nitrate and borate is recommended at startup. Protect the system from slime or organic growth.